

Copy only

Rutgers-The State University
Master's Comprehensive Examination

December 3, 1983

Theory

Part I

- Instructions:
1. Open book and open notes
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded

-
1. A box contains three \$1.00 bills and two \$2.00 bills. A coin is tossed. If a head appears one bill is selected, at random, from the box. If a tail appears, two bills are selected, at random, from the box. Let X = the total money taken from the box. Find

- a) $E\{X\}$.
- b) $\text{Var}(X)$.
- c) $P(\text{coin toss resulted in heads} \mid X = \$2.00)$.

2. a) Let $X_i, i=1 \dots n_1$ have distribution $N(\mu_{X_1}, \sigma_{X_1}^2)$

Let $Y_j, j=1 \dots n_2$ have distribution $N(\mu_{Y_1}, \sigma_{Y_1}^2)$
all random variables are independent. Find $P[\bar{X} > \bar{Y}]$, Give the numerical value of this probability for $n_1=10, n_2=15, \mu_X = \mu_Y$ and $\sigma_X^2 = \sigma_Y^2 = 6$.

- b) Let X_1, X_2 be independent r.v. distributed as negative exponential, with parameter $\theta = 1$, derive the density of $\sqrt{X_1 + X_2}$
3. Let $X_1, \dots, X_{10}$ be i.i.d. uniform random variables on $(0, \theta)$. Find an upper confidence bound for θ with confidence coefficient .95

4. Let $X_1, \dots, X_n$ be i.i.d. with a density

$$f(x) = \theta e^{-\theta(x-\phi)}, \quad x > \phi, \quad \theta > 0, \quad \phi \text{ known.}$$

- (i) Consider testing $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$ at level α let the critical region be $\bar{X} < c_\alpha$ for some constant c_α , show this to be a UMP level α test.
- (ii) For large n using central limit theorem find an approximate value for the critical region for $\alpha = .05$.
- (iii) Using normal approximation find the smallest value of n required to have a power at least .95 for all alternatives $\theta > 1.1 \theta_0$ when the level of test is .01.

5. Consider a random sample of size n from a distribution with density

$f_x(x|\theta) = \theta_1 \theta_2^{\theta_1} / x^{\theta_1+1}$, $x > \theta_2$, $\theta_1, \theta_2 > 0$. For θ_1 known, find a sufficient statistic, the maximum likelihood estimator, and an unbiased estimator of θ_2 .

6. Let $X_1, \dots, X_n$ be i.i.d $N(\theta, 1)$

- (i) Find the m.l.e. for θ^2 . Is it unbiased? What is its asymptotic distribution?
- (ii) Find an unbiased estimator for θ^2 .
- (iii) Compare the m.s.e's for the estimators you find in (i) and (ii).

Rutgers-The State University
Master's Comprehensive Examination

Part II

Applied

December 2, 1983

- Instructions:
1. Open book and open notes
 2. Do four out of the following five problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded
-

1. In 1983 of the 2000 Freshmen entering Rutgers College 10% had SAT scores over 700 and 35% had SAT scores over 600. In 1982 comparable figures were 12% and 36%. On the basis of this information test whether the quality of incoming freshmen is changing. Describe in quantitative terms any changes you find.
2. To test a method for the quantitative analysis of calcium oxide, known quantities X in milligrams were analyzed. With one exception (starred) all gave under estimates of the known content. Determine the regression of Y on X and whether it has a zero intercept. Test whether this starred result can be considered an outlier and at what probability.

Taken X	16.0	20.0	25.0	31.0	36.0
Found	15.6	19.8	24.5	31.1*	35.5

3. The effect of 3 diodes on current are to be studied. As the effects might depend on temperature, observations are to be taken on each diode at each of 3 temperatures. The temperature is set and each diode is tested in random order.

The results are

	diode 1	diode 2	diode 3
Temp 1	4.0	4.8	5.0
2	4.4	5.0	5.2
3	4.8	4.6	5.2

- a) What type of design is this
- b) Calculate the ANOVA table, what model are you fitting? What do you conclude? Find the EMS.
- c) If it was known from previous experiments that the variance of the measurements is approximately .1 what would you conclude?

4. The following data is sales of a liquid chemical.
Sales for 200 days
(Thousands of gallons)

	# of days
less than 34	0
34.0 and under 37.0	33
37.0 and under 40.0	78
40.0 and under 43.0	78
43.0 and under 46.0	11
46.0 or more	<u>0</u>
	200

- a) At the 1% level perform a test that sales are normally distributed.
b) What other indications do you have that the data is or is not reasonably close to normal?
5. The following figures are behavior ratings before and after 10 weeks of receiving a placebo, for 10 chronic schizophrenics.
- a) Use two different nonparametric tests to test the null hypothesis of no effect of this drug. Given $\alpha = 5\%$. What do you conclude? Explain any seeming contradictions.
- b) Using nonparametric procedures estimate the placebo effect and give a 95% confidence interval (as close to 95% as possible)
- Before 2.5, 3.41, 2.2, 1.9, 2.85, 2.64, 3.1, 4.1, 2.9, 3.03
- After 1.9, 3.6, 2.5, 3.1, 2.80, 3.05, 3.32, 4.16, 3.16, 3.15

Rutgers-The State University
Master's Comprehensive Examination

March 12, 1983

Part I

Theory

- Instructions:
1. Open book and open notes
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded

-
1. A fair die is tossed n times. Let X_i denote the outcome of the i th toss. Use a well-known inequality to find a lower bound for

$$P(3n < \sum_{i=1}^n X_i < 4n).$$

2. Assume X_1, X_2 are iid $N(0, \sigma^2)$ random variables. Compute

$$P\left[\frac{(X_1 + X_2)^2}{(X_1 - X_2)^2} < 1\right].$$

Hint: Note $(X_1 - X_2)^2 = (X_1 - \frac{X_1 + X_2}{2})^2 + (X_2 - \frac{X_1 + X_2}{2})^2$

3. Suppose X has a Gamma $(\frac{n}{2}, 2\theta)$ distribution and we have a Gamma (α, β) prior on $\frac{1}{\theta}$.

- (i) Find the form of the posterior
- (ii) What is the Bayes estimator of θ under squared error loss?

Note: The Gamma (α, β) density is: $f(x|\alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, x > 0$.

The Inverse Gamma (α, β) density is:

$$g(x|\alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha x^{\alpha+1}} e^{-1/x\beta}, x > 0.$$

4. Let $X_1, \dots, X_n$ be a sample from a distribution with density

$$\begin{cases} \frac{1}{\sigma} \exp(-(x-\mu)/\sigma) & \text{if } x > \mu \\ 0 & \text{otherwise} \end{cases}$$

Assume $\sigma > 0$ is known and $-\infty < \mu < \infty$.

- (i) Find a sufficient statistic.
- (ii) Find a MLE for μ .

5. Let $X_1, \dots, X_n$ be a sample from a Poisson (λ) distribution. We want to estimate $\theta = P_\lambda(X=0)$, using squared error loss. Consider the estimator

$$d_1(X_1, \dots, X_n) = \begin{cases} 1 & \text{if } X_1 = 0 \\ 0 & \text{if } X_1 > 0 \end{cases}$$

Note that d_1 is an unbiased estimator of θ . Using sufficiency (Rao-Blackwell theorem) find an unbiased estimator of θ , which is better than d_1 .

6. We are interested in finding information about the precision of a measuring instrument. We use the instrument on a known standard and assume that the observations (measurements) are normally distributed with known mean μ_0 and unknown variance σ^2 . The hypothesis of interest is $H_0: \sigma \leq \sigma_0$ and the alternative $H_a: \sigma > \sigma_0$.

- (i) Show that the UMP level α test for H_0 vs H_a has rejection region

$$R = \left\{ X: \sum_{i=1}^n (X_i - \mu_0)^2 > \sigma_0^2 \chi_n^2(1-\alpha) \right\}$$

where $\chi_n^2(1-\alpha)$ is the $1-\alpha$ quantile of the χ_n^2 distribution.

- (ii) Using chi-square tables find an approximation to the smallest sample size required in order to have power at least .9 at level .01, against all alternatives with $\sigma > 1.1\sigma_0$.

Rutgers-The State University
Master's Comprehensive Examination

March 12, 1983

Part II

Applied

- Instructions:
1. Open book and open notes
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded

1. Subjects in both the 1973 and 1974 General Social Survey were asked how much confidence they had in the U. S. Supreme Court. The contingency table of responses is shown below.

Response	Year	
	1973	1974
A great deal	471	492
Only some	745	710
Hardly any	231	213
Don't know	50	67
No answer	7	2
Total	1504	1484

Based on these data, does there appear to be a change in attitude toward the Court? If so, describe the nature of this change. Whenever a formal test or confidence interval is used in your analysis, state the assumptions made to justify its use. The General Social Survey is an opinion survey of a national probability sample. Each year an independent sample is taken.

2. If an antibiotic is stored at high temperature it eventually loses its potency. The following data come from study to evaluate the stability of a particular antibiotic.

Days of Storage at 32 C.	Potency (in standard units)
0	623, 632
30	614, 607
60	596, 600
90	592, 582

- (a) Fit a straight line to the data.
- (b) Calculate a 95% confidence interval for the rate of loss in potency per day.
- (c) Test for lack of fit.

The following may be helpful.

$$\sum x_i = 360$$

$$\sum y_i = 4,846$$

$$\sum x_i y_i = 214050$$

$$\sum x_i^2 = 25200$$

$$\sum y_i^2 = 2,937,402$$

Master's Comprehensive Examination

3. Light bulbs of 4 different brands were left on until they went out; the survival values are as shown.

<u>Observation</u>	<u>Brand 1</u>	<u>Brand 2</u>	<u>Brand 3</u>	<u>Brand 4</u>
1	22	24	22	38
2	24	24	25	33
3	25	28	24	36
4	21	24	25	35
5	22	23	22	32
6	19	25	21	40
ΣX_i	133	148	139	214
ΣX_i^2	2971	3666	3235	7678

- (a) Is this a fixed or random design, why? State a model for this data.
- (b) Do a one-way ANOVA on the data using the a-priori contrasts

$$0 \ 0 \ 1 \ -1 \ c_1$$

$$1 \ 1 \ -1 \ -1 \ c_2$$

$$1 \ -1 \ 0 \ 0 \ c_3$$

- (c) Test the hypothesis that the 4th brand is different than the other 3 brands - assume the hypothesis occurs to the investigator after looking at the data.
- (d) Whenever a formal test is used state the assumption made to justify its use.

4. The possible association of various behavioral disorders with hormonal imbalance has long been of interest. Margolese, in a 1970 study (HORM & BEH 1, 151-155) examined the androgen to estrogen ratio in homosexual and heterosexual men. His results are indicated below:

HOMOSEXUALITY AND ENDOCRINE CORRELATE
Values of Androsterone (A), Etiocholanolone (E),
and A/E Ratios

Subject	Age	Urinary steroids, mg/24 hrs		
		A	E	A/E
Heterosexual males ("healthy")				
1	20	3.9	1.8	2.2
2	25	4.0	2.3	1.7
3	29	3.8	2.3	1.7
4	24	3.9	2.5	1.6
5	21	2.9	1.3	2.2
6	37	3.2	1.7	1.9
7	21	4.6	3.4	1.4
8	33	4.3	3.1	1.4
9	28	3.1	1.8	1.7
10	42	2.7	1.5	1.8
11	37	2.3	1.4	1.6
Homosexual males ("healthy")				
12	41	2.5	2.1	1.2
13	28	1.6	1.1	1.4
14	37	3.9	3.9	1.0
15	31	3.4	3.6	0.9
16	33	2.3	2.5	0.9
17	29	1.6	1.7	0.9
18	25	2.5	2.9	0.9
19	32	3.4	4.0	0.8
20	40	1.6	1.9	0.8
21	25	4.3	5.3	0.8
22	24	2.0	2.7	0.7
23	29	1.8	3.6	0.5
24	43	2.2	4.1	0.5
25	34	3.1	5.2	0.6
26	42	1.3	4.0	0.3

Summarize and analyze the data. Substantial numerical calculations are neither expected nor desired. Be sure to identify all assumptions being made, as well as procedures you might use for testing the validity of the assumptions. Also describe any other analyses you might wish to perform, but cannot due to time constraints. Comment on the possible limitations and conclusions from the study.

5. The following data is from a 3^2 factorial study for discrete level factors (low, middle, high)

	Data - 353			Approx. Row Averages	
	96	60	-27	129	43
	56	5	-62	-1	0
	-12	-75	-41	-128	-43
Σc	140	-10	-130	0	
Approx. column averages	47	-3	-43		

- (a) State the simplest additive model and find its fitted value and residuals.
- (b) What can you conclude from the above analysis?
- (c) Find the MSE and the mean square for the linear row by column interaction. Why is it probably not valid to test the hypothesis that this interaction is zero?
6. In a study of the comparative tensile strength of tape-closed and sutured wounds, the following results were obtained on 10 rats, 40 days after incisions made on the backs had been closed by suture or by surgical tape.

Rat	1	2	3	4	5	6	7	8	9	10
Tape	659	984	397	574	447	479	676	761	647	577
Suture	452	587	460	787	351	277	234	516	577	513

Test (at $\alpha = .05$) for no difference against the alternative that tape closed wounds are stronger using

- (a) The Wilcoxon sign test (use the normal approximation even though $N = 10$).
- (b) The sign test.
- (c) How do you account for the difference between the analysis in (a) and (b).